

Methylammonium Iodide (MAI) Enhanced, Solution Processed High-Performance Photodetector Based on Lead Sulfide Quantum Dots

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Abstract

Lead sulfide (PbS) quantum dots (QDs) have potential for next generation of photoelectrical material applied on photodetectors. A technique is demonstrated that enhances responsivity of photoelectric lead sulfide quantum dots by two step ligand exchange. The result we obtained has potential to optimize PbS QDs-based photodetectors.

Author Keywords

Quantum dots; Photodetectors; Ligand exchange; Layer by layer deposition.

1. Introduction

Photodetector is one of the most important devices in both our daily lives and research areas. It is widely used in imaging, sensing, optical communication and almost all kinds of optoelectrical instruments [1-4]. It works as a converter, transfers optical information which is hard to be calculated into electrical signal that could be easily analyzed. Currently commercial photodetectors based on gallium phosphide (GaP), silicon (Si), and indium gallium arsenide/germanium (InGaAs/Ge) are used to detect ultraviolet (UV, 200 nm ~ 400 nm), visible (400 nm ~ 750 nm) and near-infrared (IR, 750 nm ~ 2000 nm) region, respectively. For farther infrared, lead sulfide (PbS) is the most common choice. Though these devices are sensitive and well developed, most of these photodetectors are fabricated using epitaxy-grown method, which requires vacuum and expensive instruments. Another drawback is, they suffered from narrow detection range. When a large range of spectrum is required, multiple photodetectors should be combined and work together, this would surely increase the cost and the difficulty of operation.

PbS quantum dots (QDs) have broad, adjustable absorption spectrum from 400 nm to 2000 nm [5], contain wide range including infrared, visible and UV spectra, and this provided a solution for multi-spectral detection photodetectors. The broad absorption is benefit from the wide tunable band gap of the PbS QDs [6], this enables the fabrication of applications that requires special detection range. Moreover, the multiple exciton generation phenomenon observed in PbS material could provide enhanced efficiency for the devices [7-9]. There are numbers of methods to synthesis PbS QDs, and all of them involves different kinds of long ligands (typically, oleic acid(OA)) attached on the QDs' surface to promise the stability. But these long ligands separate the QDs and inhibit the electrical carriers (electrons or holes) transfer between QDs and result in a lower mobility [10,11].

In the last decade, scientists have make efforts on making efficient photodetectors using PbS QDs. Different kinds of structures are put forward to enhance the device performance. The photoconductors [12] have the most simple and easy fabricated structure. Compared with simple photoconductors, phototransistors [13] and photodiodes [14] enhanced the efficiency of the devices dramatically, but the cost is also increased. Metal nanoparticles such as Au and Ag have also

been used by doping into PbS QDs to achieve a better photocurrent, but this also produce a drawback of the response time [15]. Another direct solution is exchanging the long ligands with short ligands such as hydroxide ions (OH⁻), sulfide (S²⁻) and iodide (I⁻) [10,16-19]. These short ligands decreased the gap between QDs and enhanced the mobility. The exchange process is completed by a simple drop and reaction step. This easy accessed method is cost efficient, but the long ligands may not be completely exchanged and will remain on the QDs. MAI pretreatment has been used in photovoltaic to enhance the ligand exchange ratio [20], but it has never been applied on photodetectors. Moreover, some materials that could enhance the electron transmission, such as graphene and ethanedithiol (EDT) [21], is also combined with PbS QDs photodetectors (in which the holes are the majority carrier) to optimize the device performance.

In this paper, a MAI enhanced, solution processed PbS QDs photodetector with ultra-responsivity is demonstrated. The OA long ligands are replaced with iodide ions (I⁻), moreover, layer-by-layer (LBL) spin-coating is also applied to reach an optimized mobility. The PbS QDs are MAI treated before the device fabrication process to reach a higher exchange ratio, the concentration of MAI is considered and designed in the experiment. The absorption, PL intensity, surface morphology, I-V, detectivity and responsivity are characterized to evaluate the material and the devices. All devices show expanded detection region from 400 to 1000 nm. And for the best performance device, an enhancement of 67.1% is achieved using 0.01M MAI pretreatment. The optimized responsivity reaches 2.34 A/W. The PbS QDs photodetectors show its potential of becoming the solution of multi-spectra detectable photodetectors.

2. Result and Discussion

The photodetector is mainly evaluated by the following four parameters:

Detection range is decided by the absorption spectrum of the material, as is mentioned above, a wide detection range is better for basic devices;

Responsivity (R) is defined by the generated photocurrent and incident light power ratio, could be calculated using equation (1):

$$R(\lambda) = \frac{I_{ph}(\lambda)}{P_{in}(\lambda)} = \frac{I_p(\lambda) - I_d(\lambda)}{P_{in}(\lambda)} \quad (1)$$

Where I_p refers to photocurrent under illuminance, I_d refers to dark current.

Responsivity interrelates with external quantum efficiency EQE(λ) and photoconductive gain G(λ) as:

Where h refers to Plank' constant, c refers to the speed of light, q refers to the elementary charge;

$$R(\lambda) = \frac{EQE(\lambda)G(\lambda)}{1} \cdot \frac{q\lambda}{hc} \quad (2)$$

Detectivity (D^*) is used to value the device performance, considered the active area of the devices, could be calculated through responsivity, noise current spectral density (I_n) and the active area (A) using equation (3).

$$D^*(\lambda) = \frac{R(\lambda)}{\sqrt{I_n^2}} \cdot \sqrt{A} \quad (3)$$

Noise current in a photodetector consist of low frequency flicker noise, thermal noise and shot noise. In our case, with approximation, the equation (3) is simplified as:

$$D^*(\lambda) = R(\lambda) \cdot \sqrt{\frac{A}{2qI_d}} \quad (4)$$

Bandwidth of a photodetector is a key issue for communication applications. It decided how much information the photodetector could accept in a period of time.

As is mentioned above, the shorter ligands attached on QDs could enlarge the mobility of the QDs film. And this increasement of mobility would product a higher photoconductive gain $G(\lambda)$, which results in an enhancement of responsivity.

The device is designed as a simple photoconductor structure (figure 2. (a)), the QDs are covered on a patterned transparent ITO electrode using a LBL spin-coating method.

Absorbance and PL spectra

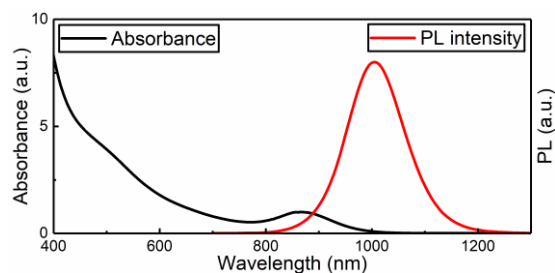


Figure 1. The absorption and PL spectra of synthesized PbS QDs

The absorption spectrum shows a wide absorption from 400 nm to 1000 nm, with a peak absorption at 860 nm. The photoluminescence (PL) intensity with a single peak at 1000 nm shows the uniformity of the synthesized QDs. The absorbance and PL could be adjusted by controlling the size of QDs during the fabrication process.

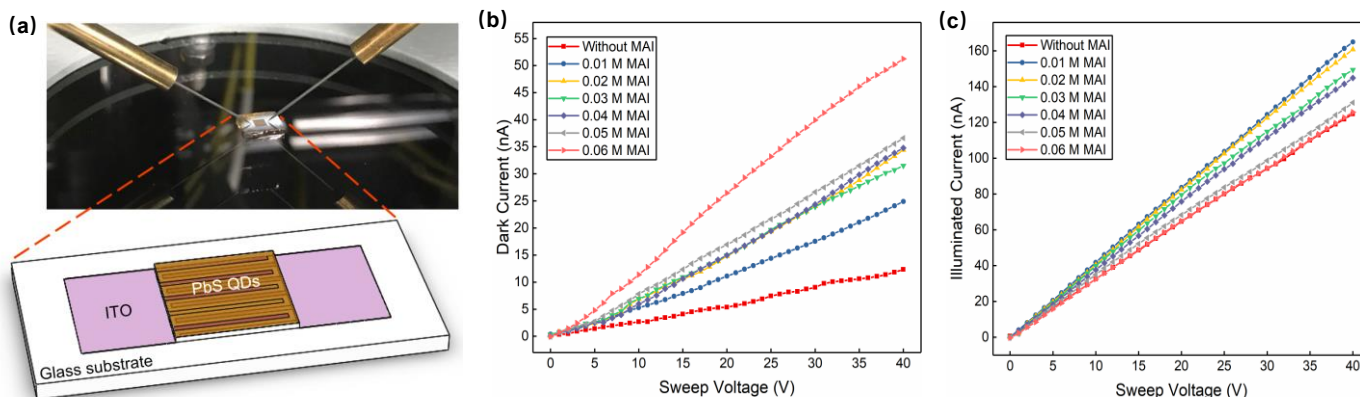


Figure 2. The testing set up and the designed device structure (a), and the dark (b) and illuminated current (c) for all devices. The light source is a green light at 550 nm with intensity of 110 uW/cm².

Devices with different MAI concentration

To find the best amount of MAI, 7 groups of devices with different concentration of MAI are prepared using exactly the same method.

The dark current shows the conductivity enhancement by using different concentration of MAI. Apparently, high concentration MAI treated devices have higher dark current compared with untreated device. Higher concentration of MAI surely brings more short ligands onto PbS QDs and result in an increasing dark current with more amount of MAI.

Illuminated current gives a different trend with dark current. Enhancement is observed in all MAI treated devices except for the highest 0.06 M one. And the largest illuminated current shows when MAI is only at 0.01 M. After 0.01 M, excess MAI lower down the current dramatically. The enhancement of illuminated current could also be considered as a result of increased mobility, in another word, short ligands attached on PbS QDs. But in high concentration, the increased mobility does not dominate the illuminated current any more.

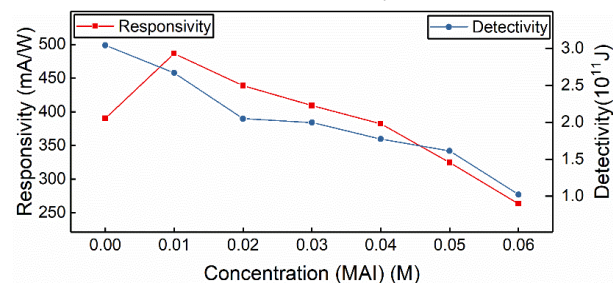


Figure 3. The responsivity (R) and detectivity (D^*) for the devices achieved above, under 550nm, 110uW/cm².

Using equation (1) and (4), the responsivity and detectivity of the devices are obtained (shown in Figure 3). At lower MAI concentration (less than 0.04 M), the responsivities are all enhanced, the optimized device shows 29% enhancement from 386 mA/W to 498 mA/W. The detectivity drop a little because of the increasing dark current.

Similar with the illuminated current curves, the responsivity (current) reaches the best point at 0.01 M MAI, then drop down continually. This may be caused by the irreversible precipitation of QDs when having too much amount of MAI. Though the fabrication processes have been applied gently and slowly, the irreversible precipitation of QDs is still unavoidable.

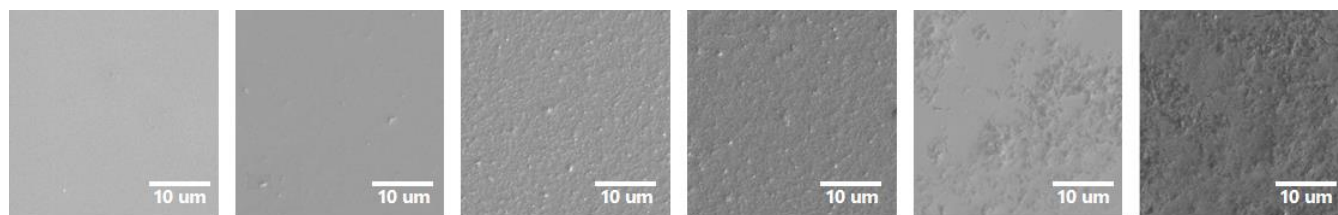


Figure 4. SEM figures for devices with 0.01 M ~ 0.06 M MAI (left to right) treated.

SEM figures show the surface morphology of different devices. From left to right, the concentration of MAI increased from 0.01 M to 0.06 M. The defects also can be observed gradually. The first two groups have almost no defects under the shown magnification. Smooth morphology indicates a great film quality of PbS QDs. While the later devices start showing pin holes and more defects. By repeated experiments, these unsmooth surfaces are formed because of the high concentration of MAI.

Responsivity under different circumstance

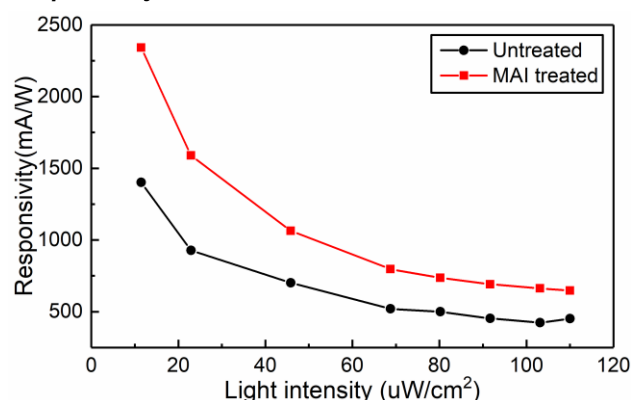


Figure 5. Responsivity at different light intensity, illuminated under a green light at 550nm.

The responsivity drops with the increasing of light intensity. And in all occasions, the MAI pretreated devices show better performance than the untreated devices. And the responsivity reaches the optimization of 2.34 A/W under 11.4 uW/cm², the enhancement from 1.40 A/w to 2.34 A/w reaches 67.1%.

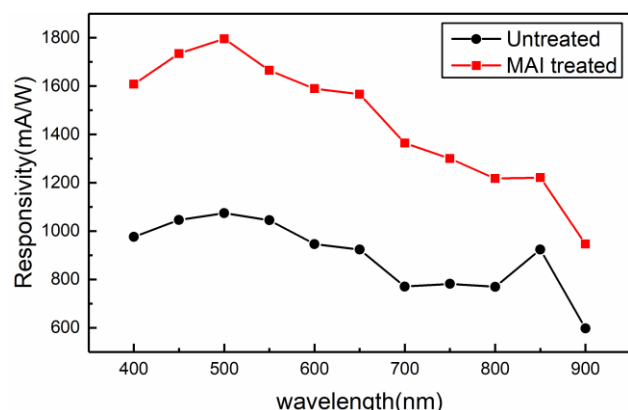


Figure 6. Responsivity of devices with or without MAI pretreatment. Illuminated under 22.9uW/cm².

Then the responsivity at different wavelength of light is obtained in figure 6. Also, the enhancement could be observed clearly. Basically, the maximum responsivity shows at 500 nm, moreover, a small peak could be observed around 850 nm

because where located the absorption peak of PbS QDs.

3. Conclusion

In summary, a high responsivity photodetector is designed and fabricated using a MAI pretreated, LBL ligand exchange process. The MAI pretreatment enhanced the device performance dramatically. While using this MAI pretreatment, the concentration of MAI should be carefully controlled, or the QDs could suffer from irreversible precipitation. The optimized responsivity reaches 2.34A/W, with an enhancement of 67.1% compared with the untreated one. Our work shows a simple and effective method of optimizing the responsivity of PbS QDs based photodetectors. The theoretical detection range of PbS QDs should be much wider than our samples, yet we only applied our method with one type of PbS QDs because of the limitation of our material and testing devices. This MAI pretreated, two-step ligand exchange method shows its potential of solution to multi-spectra, high performance PbS QDs based photodetectors.

Experiment section

PbS QDs Synthesis: The PbS QDs are prepared using a low temperature, fast injection method. 4mmol lead oxide (PbO) is added into a solution mixed with 4.4 mL OA and 25 mL octadecene (ODE) as the precursor of lead. The mixture is put into a flask and slowly heated to 90 °C, with continuous evacuation and stirring, the solution finally achieved a clear liquid under nitrogen atmosphere. The precursor of sulfide is made by dissolving 0.42mL bis(trimethylsilyl)sulfide into 12.5 mL ODE. Then the precursor of sulfide is quickly injected into the flask within 1 second, speed up the stirring, hold for 30 seconds. Withdraw the heat and take the PbS solution to further purification. The as prepared PbS QDs are purified by adding a mixture of ethanol and acetone, followed by high speed centrifugation. The extracted QDs are then dissolved in hexane and then precipitated twice with a mixture of ethanol and acetone and finally re-dispersed in octane (50 mg/ml) and toluene (100 mg/mL, for MAI treatment) respectively.

MAI treatment: For MAI treatment process, toluene-DMF (2:1 vol%) dispersed MAI (6 groups, from 0.01M to 0.06 M) solution is added to PbS QDs in toluene, as is mentioned above (the ratio of MAI solution and PbS QDs is 3:4). The treatment process is achieved drop by drop to avoid irreversible precipitation of QDs. The PbS QDs solution is then kept in the glovebox for 12 h. Methanol is used to pacificate the QDs. Following twice precipitations, the QDs are dispersed in octane with a concentration of 50 mg/mL. Then ultrasonic treatment is applied to ensure better dispersion.

Device fabrication: The device is fabricated on patterned ITO electrodes. The pattern is designed as 2 mm in length and 80 um in gap. A laser printer is used to achieve the patterned electrodes. The fabricated electrodes are then cleaned using ultrasonic cleaner, acetone, n-hexane and DI water are used step

by step to achieve a clean top surface. Drop MAI treated PbS QDs/octane solution (50 mg/mL), spin-coating at 2500 rpm for 15s, the acceleration is set as 1250 rpm/s. Drop TBAI/methanol (5 mg/mL) covering the surface and hold for 30 s, spin-off the excess solution. Drop methanol and spin-off to further clean the top surface. Wait for 60s and start the next layer

Device characterization: The absorption spectra are recorded by the Perkinmer lamda 950 spectrophotometer. The PL measurements are done with Oceanview System. The electrical characterizations are recorded on a Keithley 4200 and a SOFN 7-SCF06A/B probe station under ambient atmosphere at room temperature and the devices are illuminated by Tungsten Lamp House (7ILT250A) which can be driven transiently by the software. The electrodes are patterned by HAN'S LASER (YLP-F10). The intensity of the monochromatic light is detected by PM100D Compatible Sensors.

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